

CUAI Physical AI & VLA Study

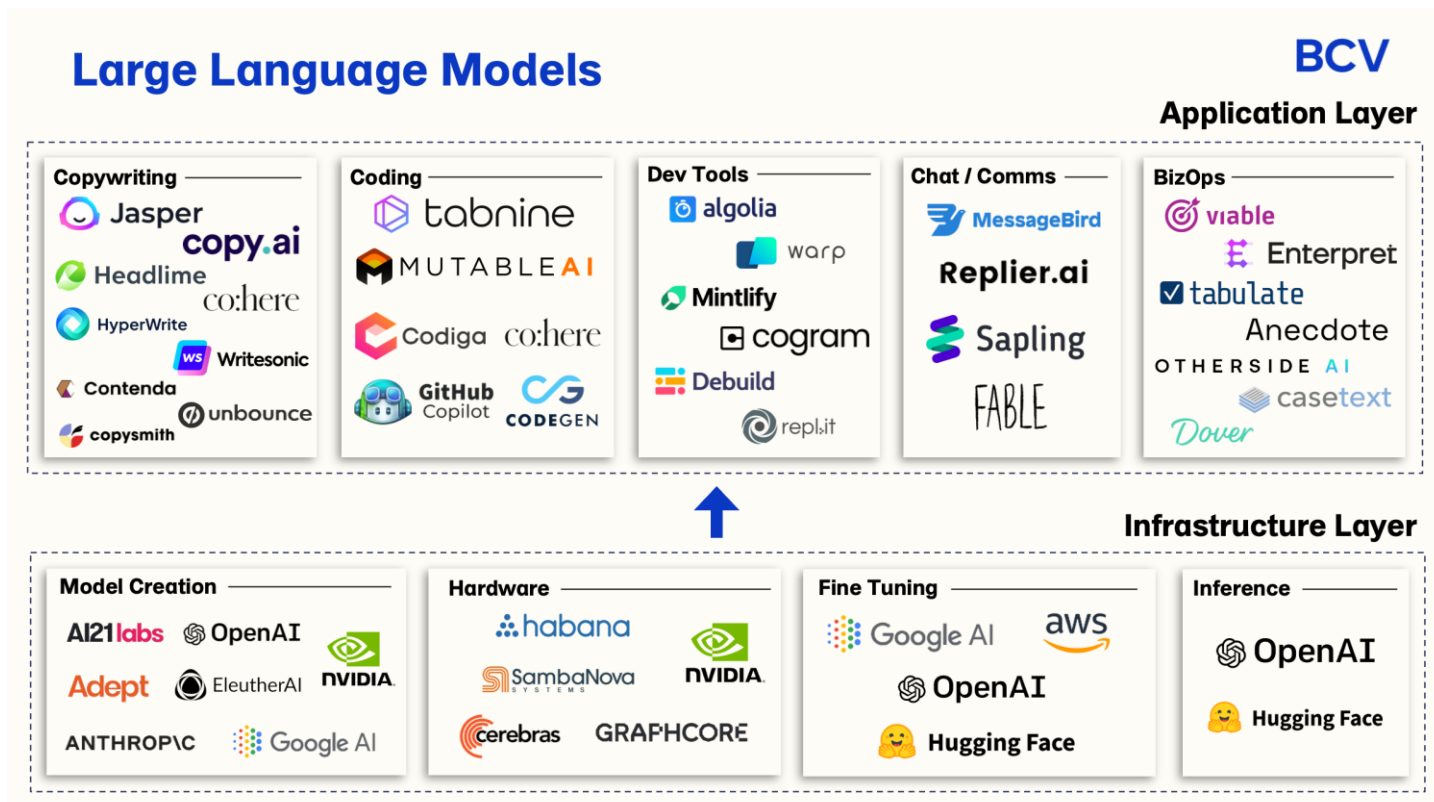
2026.09.29

발표자 : 오규안

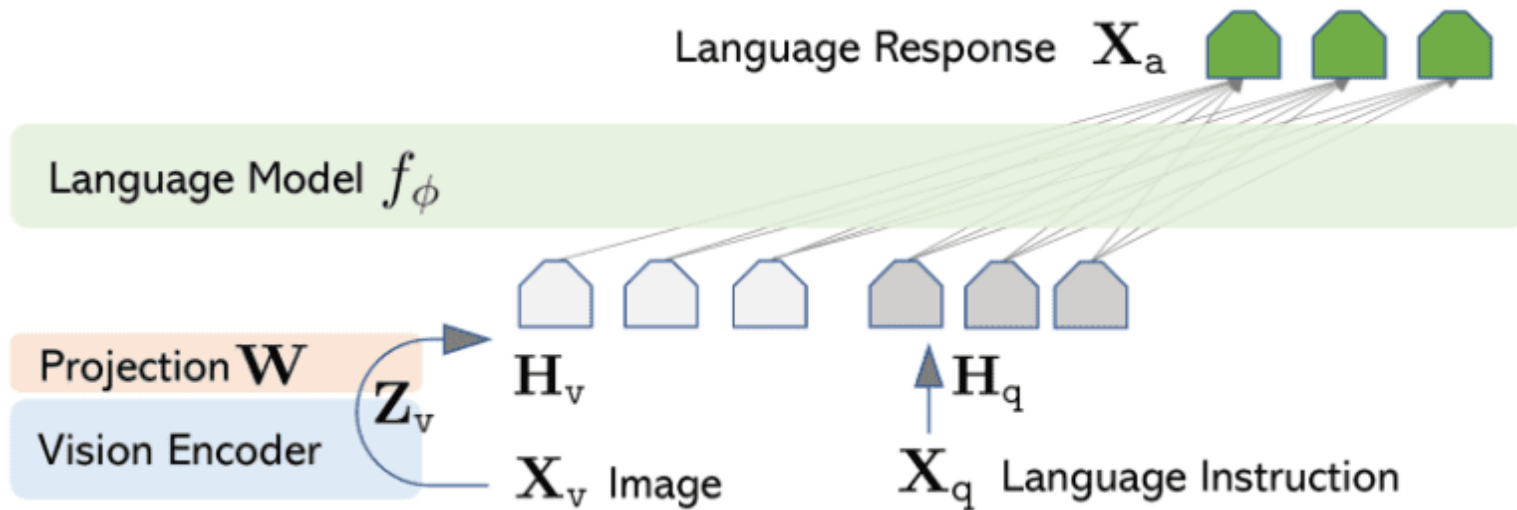
RoadMap of Physical AI



Large Language Models (LLMs)



Vision-Language Models (VLMs)



Vision + Language

- 이미지와 텍스트를 함께 이해하는 Multimodal Model

Aligning Visual and Linguistic Information

- 시각적 특징을 언어 공간과 연결해 의미를 해석

Reasoning About the Visual World

- 단순 인식을 넘어, 이미지에 대해 설명하고 질문에 답하며 추론

Vision-Language Models (VLMs) - Tasks



Instruction-based evaluation



Provide a detailed description of the given image.

The image features a **table** with a variety of food items displayed in bowls. There are two bowls of food, one containing a mix of vegetables, such as **broccoli** and **carrots**, and the other containing meat. **The bowl with vegetables** is placed closer to the front, while **the meat bowl** is situated behind it. In addition to the main dishes, there is an **apple** placed on the table, adding a touch of fruit to the meal. A **bottle** can also be seen on the table, possibly containing a **beverage** or **condiment**. The table is neatly arranged, showcasing the different food items in an appetizing manner.



POPE

Random settings



Is there a **bottle** in the image?

Yes, there is a bottle in the image.



Popular settings



Is there a **knife** in the image?

Yes, there is a knife in the image.



Adversarial settings



Is there a **pear** in the image?

Yes, there is a pear in the image.



Figure 1: Cases of object hallucination in LVLMs. **Bold** objects are ground-truth objects in the annotations and **red** objects are hallucinated objects by LVLMs. The left case is from the traditional instruction-based evaluation method, and the right cases are from three variants of POPE.

Vision-Language Models (VLMs) - Tasks

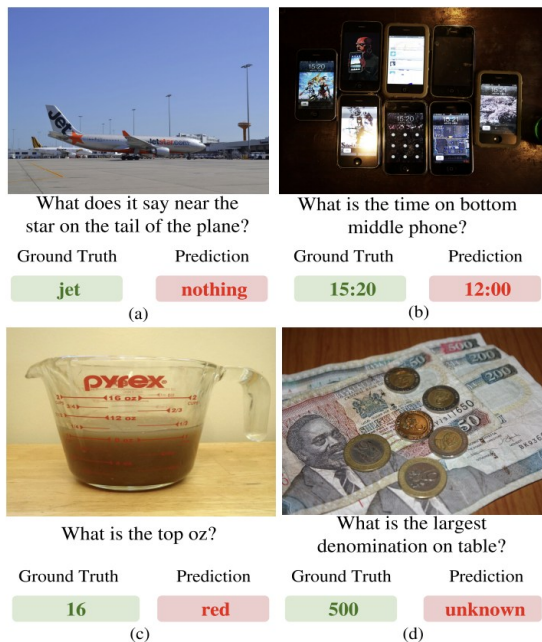
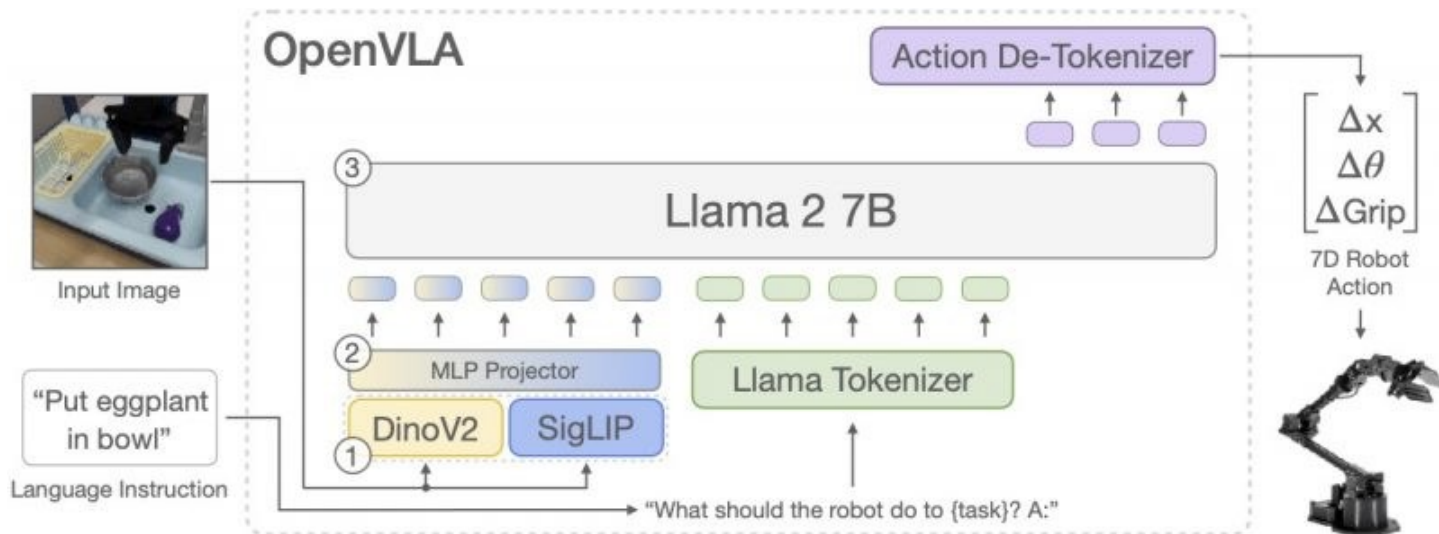


Figure 1: **Examples from our TextVQA dataset.** TextVQA questions require VQA models to understand text embedded in the images to answer them correctly. Ground truth answers are shown in green and the answers predicted by a state-of-the-art VQA model (Pythia [17]) are shown in red. Clearly, today's VQA models fail at answering questions that involve reading and reasoning about text in images.



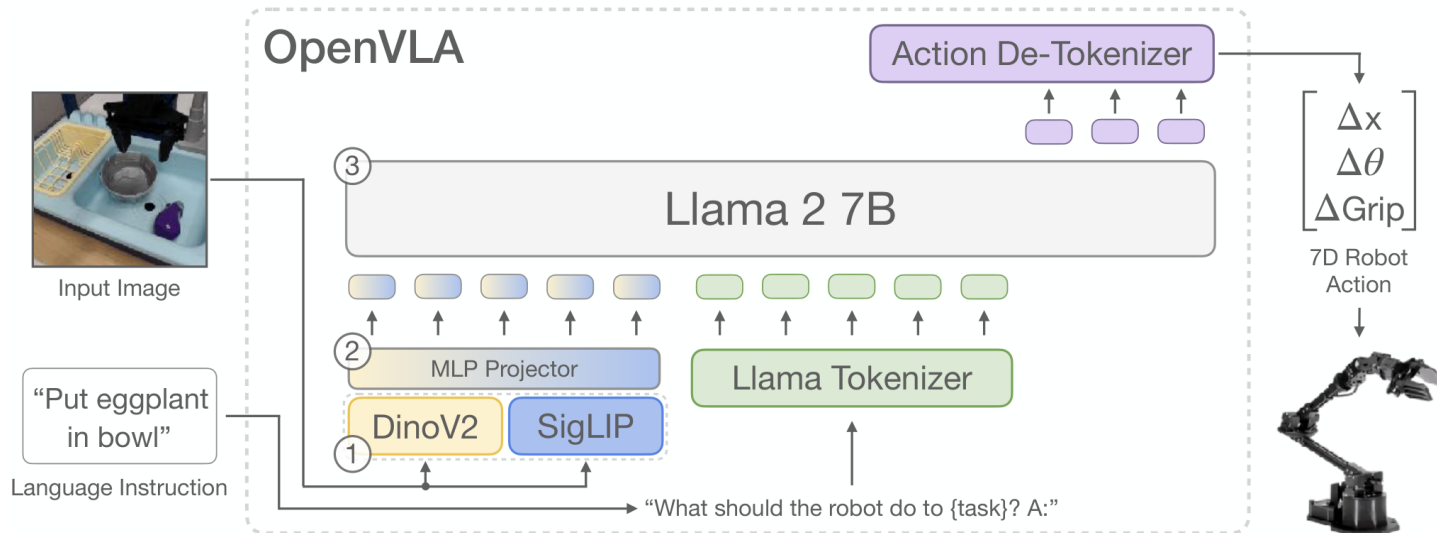
Vision-Language Action Models



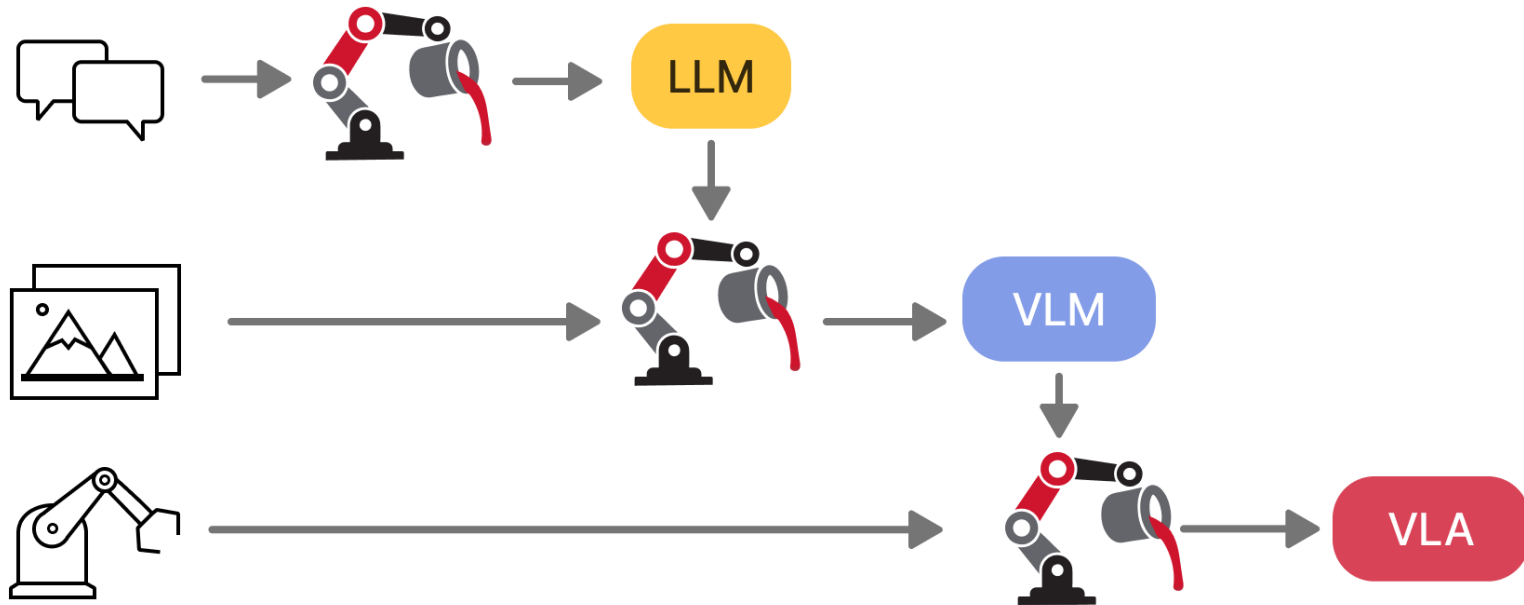
RoadMap of Physical AI



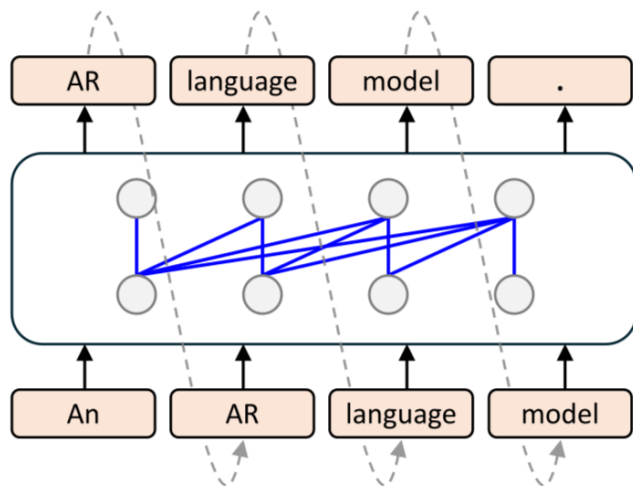
VLM to VLA



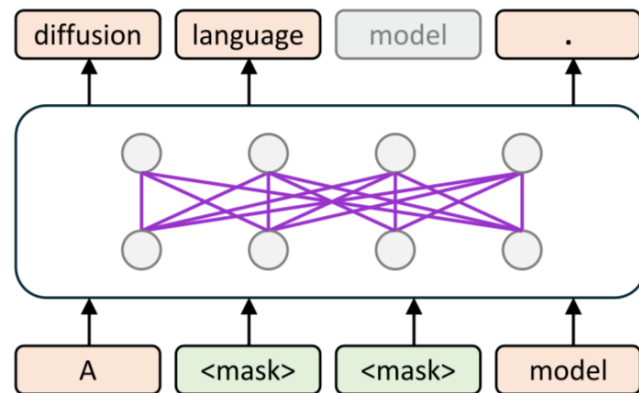
RoadMap of Physical AI



Pre) Autoregressive vs Parallel (DiffusionLLM)



(a) Autoregressive Modelling



(b) Diffusion Modelling in DREAM

OpenVLA (Autoregressive Model)

OpenVLA: An Open-Source Vision-Language-Action Model

Moo Jin Kim^{*,1} Karl Pertsch^{*,1,2} Siddharth Karamcheti^{*,1,3}
Ted Xiao⁴ Ashwin Balakrishna³ Suraj Nair³ Rafael Rafailov¹ Ethan Foster¹ Grace Lam
Pannag Sanketi⁴ Quan Vuong^{5,†} Thomas Kollar³ Benjamin Burchfiel³ Russ Tedrake^{3,6} Dorsa Sadigh¹
Sergey Levine² Percy Liang¹ Chelsea Finn¹

<https://openvla.github.io>

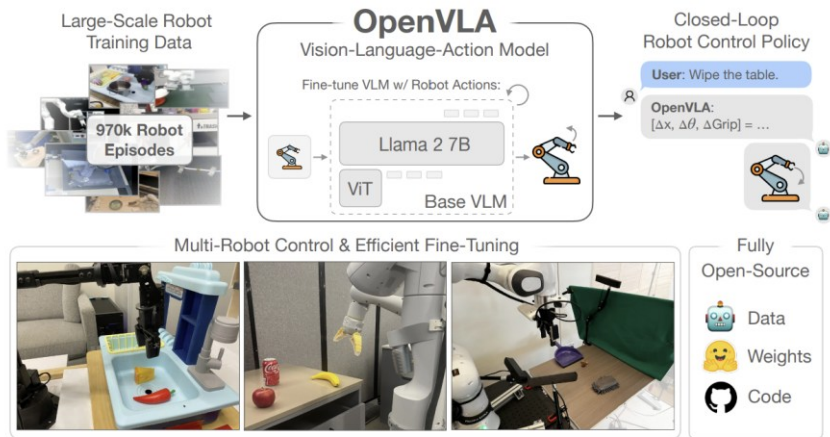


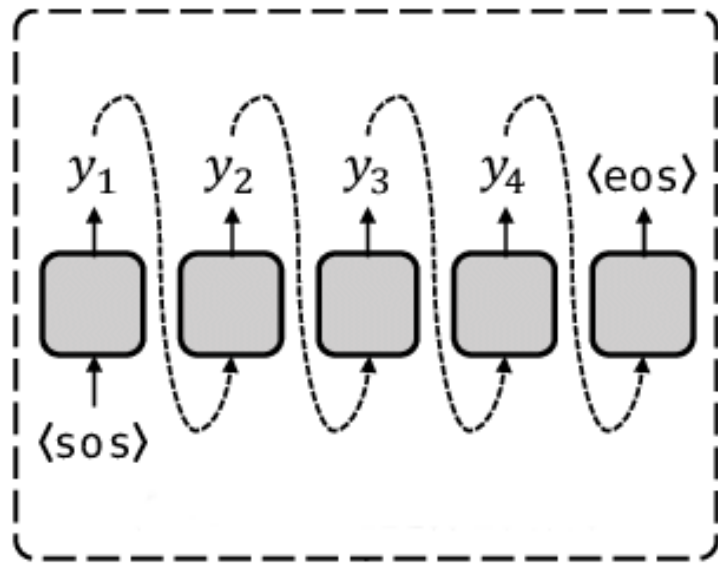
Figure 1: We present OpenVLA, a 7B-parameter open-source vision-language-action model (VLA), trained on 970k robot episodes from the Open X-Embodiment dataset [1]. OpenVLA sets a new state of the art for generalist robot manipulation policies. It supports controlling multiple robots out of the box and can be quickly adapted to new robot domains via parameter-efficient fine-tuning. The OpenVLA checkpoints and PyTorch training pipeline are fully open-source and models can be downloaded and fine-tuned from HuggingFace.

OpenVLA (Autoregressive Model)

$$p(x_i) = p(x_i | x_1, x_2, \dots, x_{i-1})$$

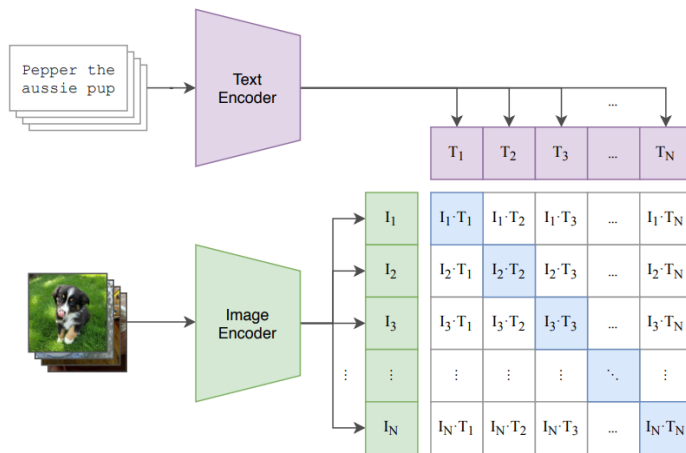
$\rightarrow$

$$p_{\theta}(x) = \prod_{i=1}^n p(x_i | x_1, x_2, \dots, x_{i-1})$$



Preliminary) SigLIP

(1) Contrastive pre-training



$$-\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\overbrace{\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_i \cdot \mathbf{y}_j}}}^{\text{image} \rightarrow \text{text softmax}} + \overbrace{\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_j \cdot \mathbf{y}_i}}}^{\text{text} \rightarrow \text{image softmax}} \right)$$

where $\mathbf{x}_i = \frac{f(I_i)}{\|f(I_i)\|_2}$ and $\mathbf{y}_i = \frac{g(T_i)}{\|g(T_i)\|_2}$. In this paper, we adopt the vision transformer architecture [17] for images and the transformer architecture [47] for texts. Note that due to the asymmetry of the softmax loss, the normalization is independently performed two times: across images and across texts [36]. The scalar t is parametrized as $\exp(t')$, where t' is a global freely learnable parameter.

Preliminary) SigLIP

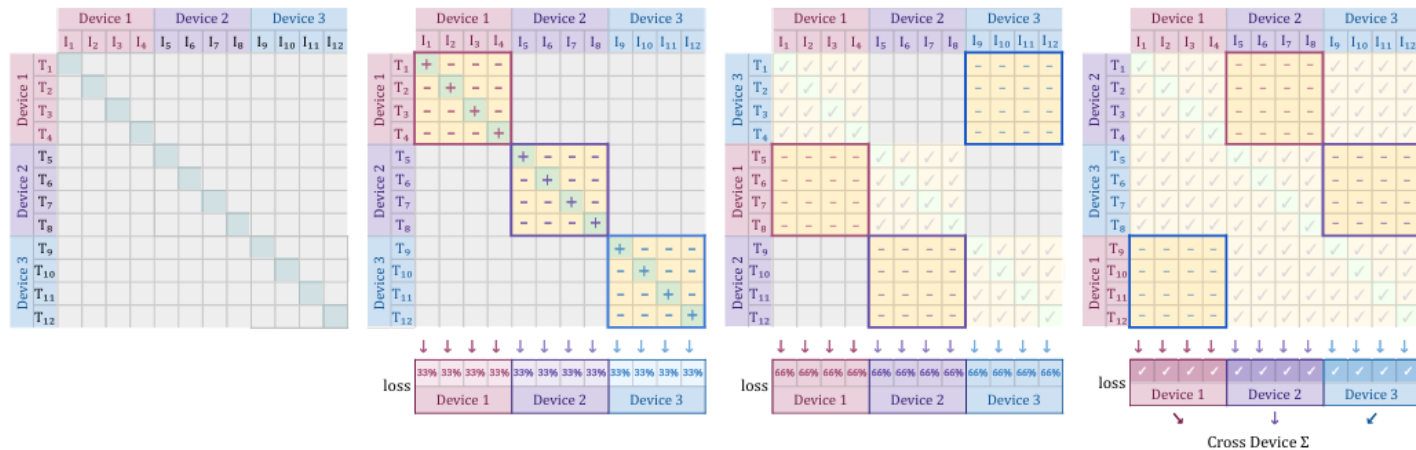
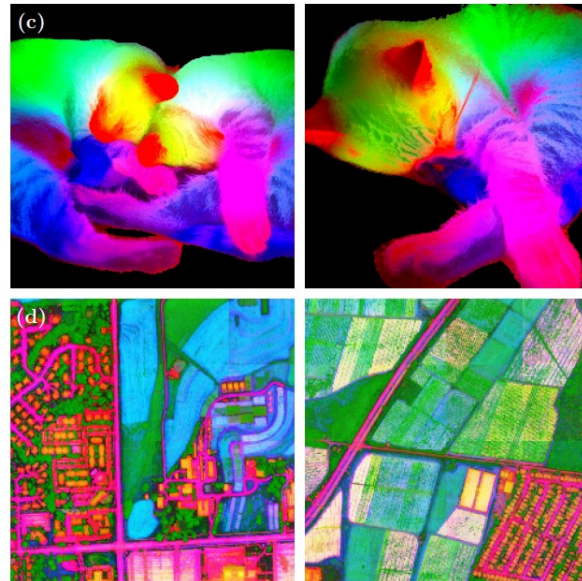
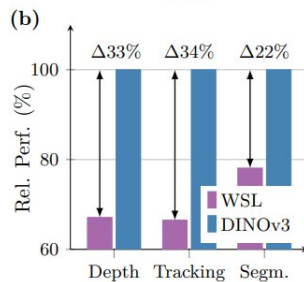
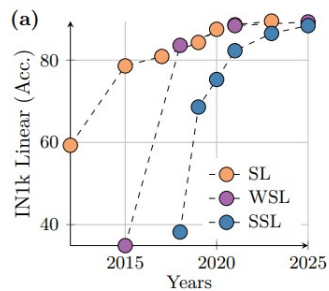
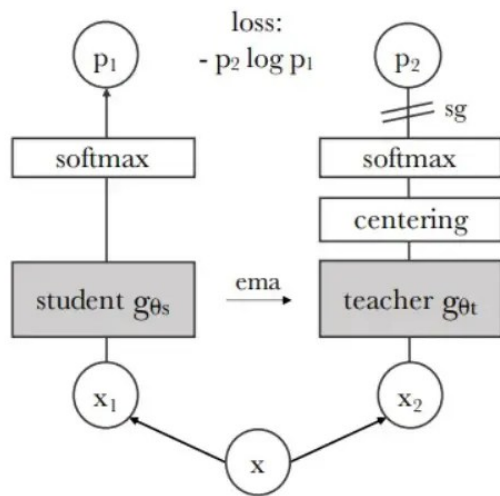
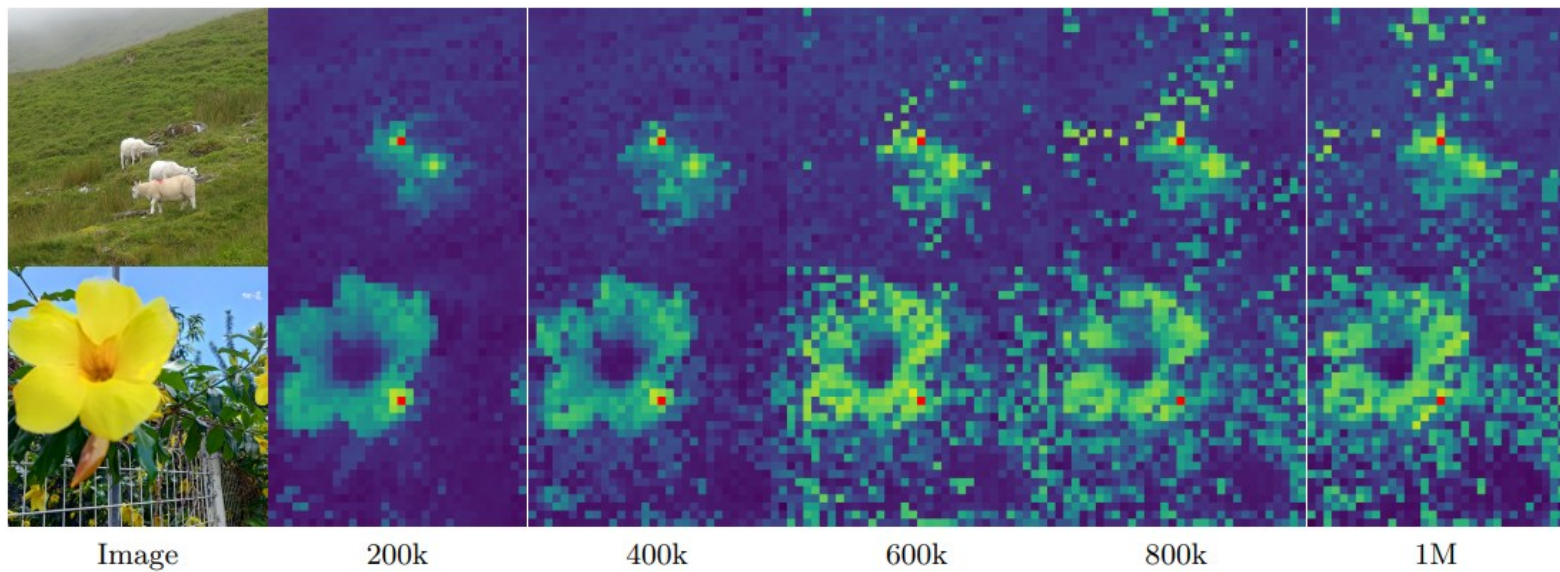


Figure 1: **Efficient loss implementation** demonstrated via a mock setup with 3 devices and a global batch size of 12. There are no all-gathers, and at any point in time only the bright yellow square (size 4×4) is materialized in memory.

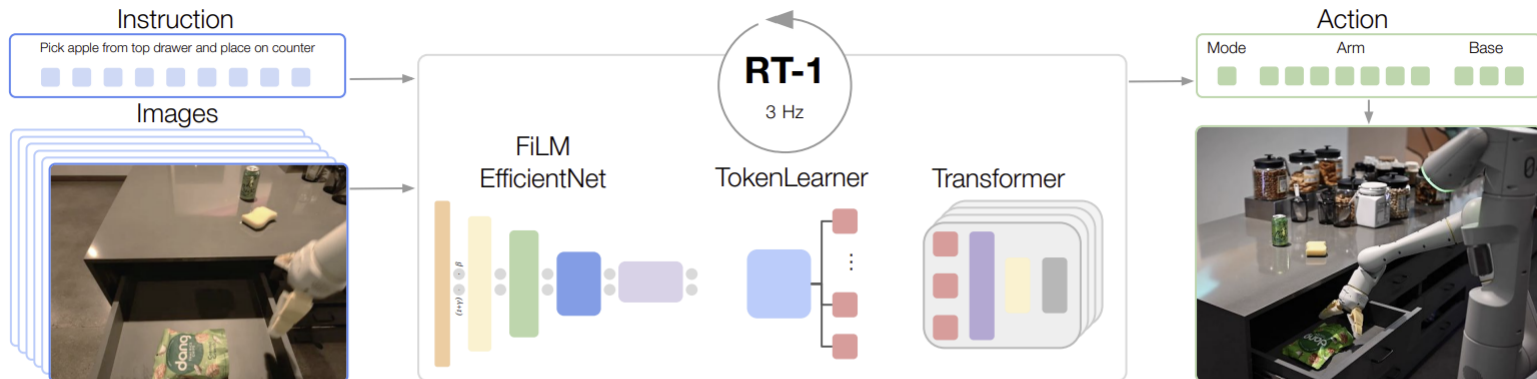
Preliminary) DINO



Preliminary) DINO

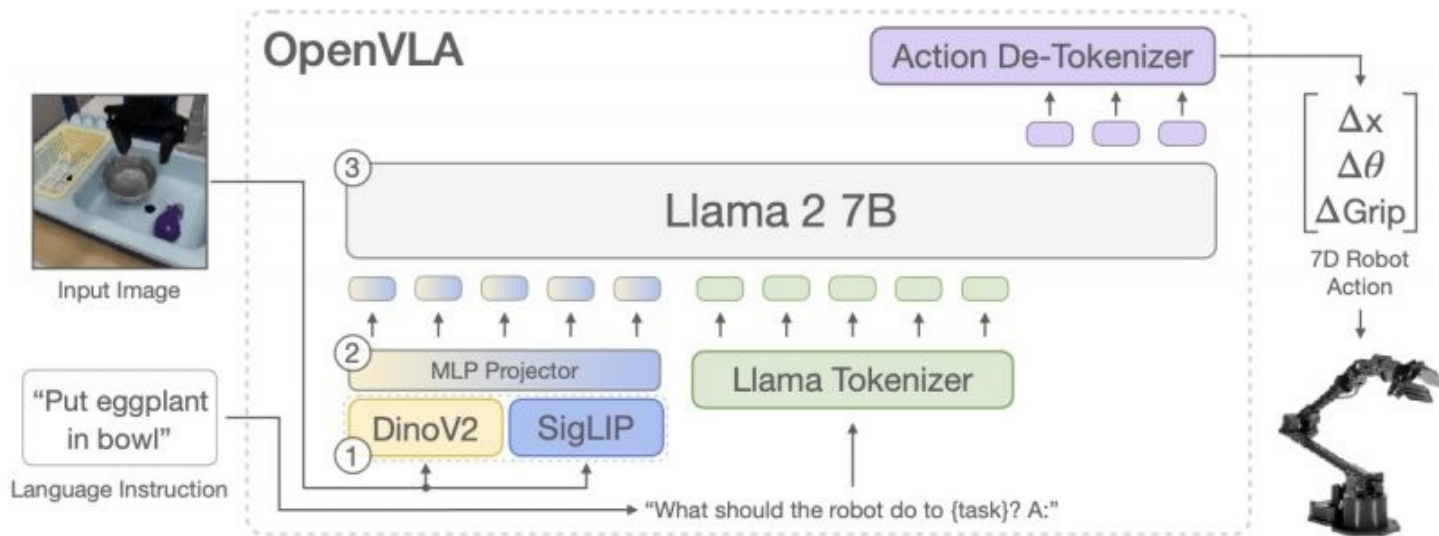


Preliminary) RT-1

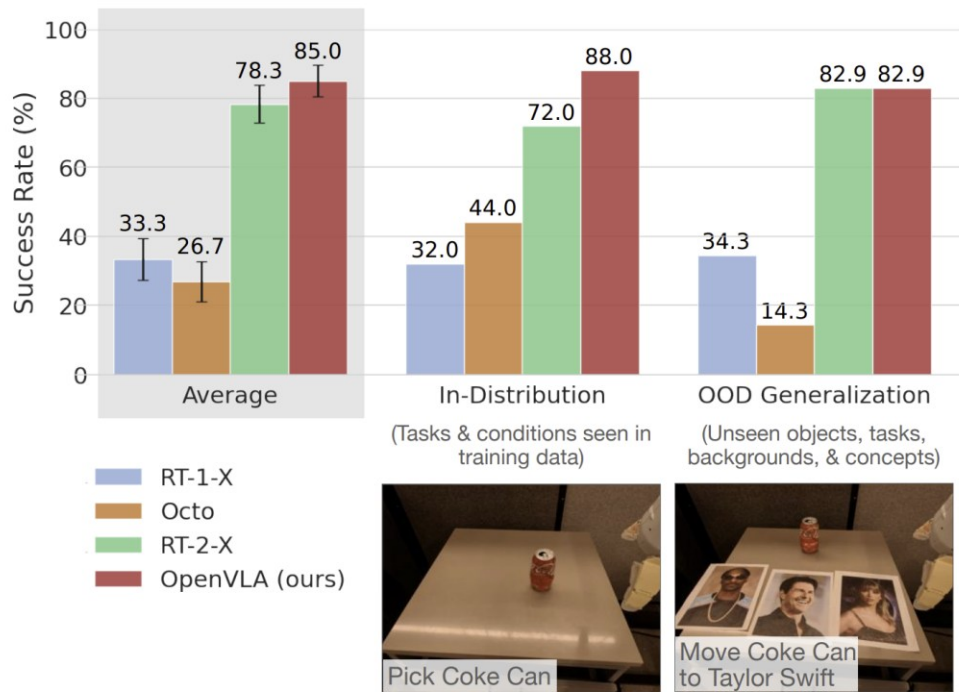


(a) RT-1 takes images and natural language instructions and outputs discretized base and arm actions. Despite its size (35M parameters), it does this at 3 Hz, due to its efficient yet high-capacity architecture: a FiLM (Perez et al., 2018) conditioned EfficientNet (Tan & Le, 2019), a TokenLearner (Ryoo et al., 2021), and a Transformer (Vaswani et al., 2017).

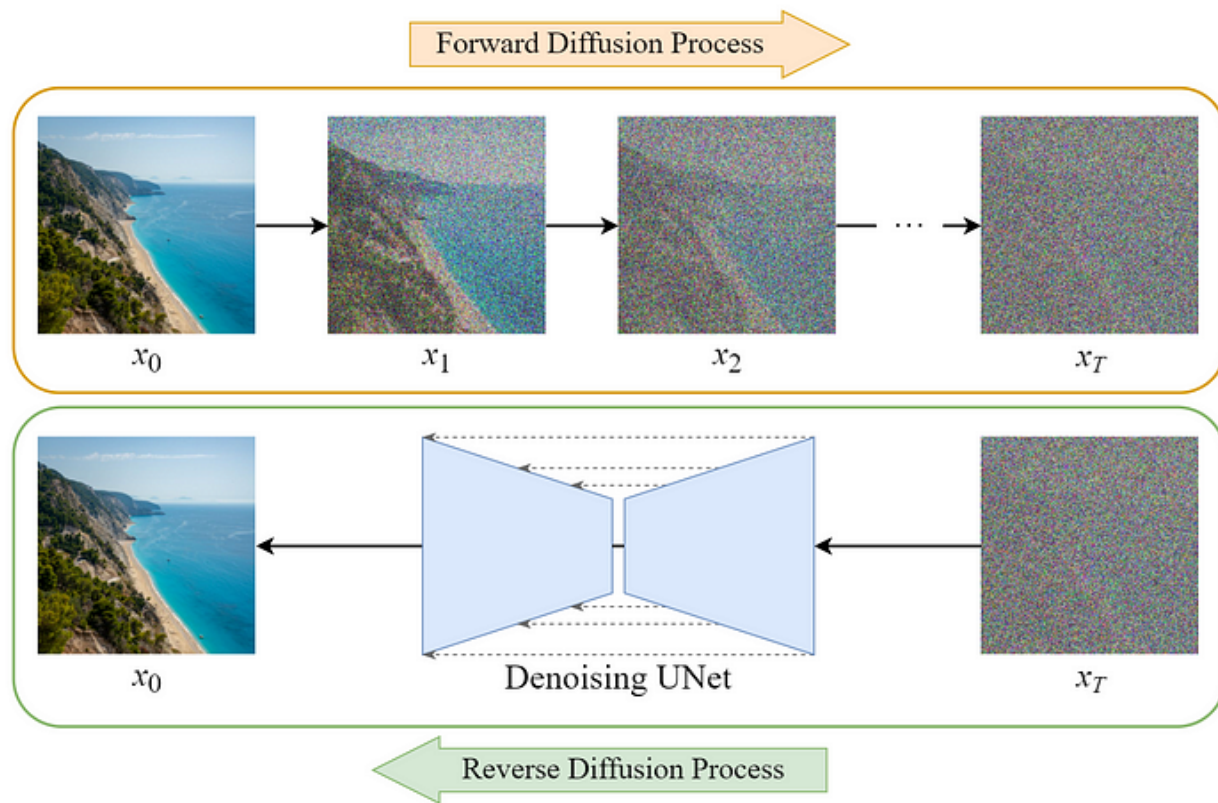
OpenVLA



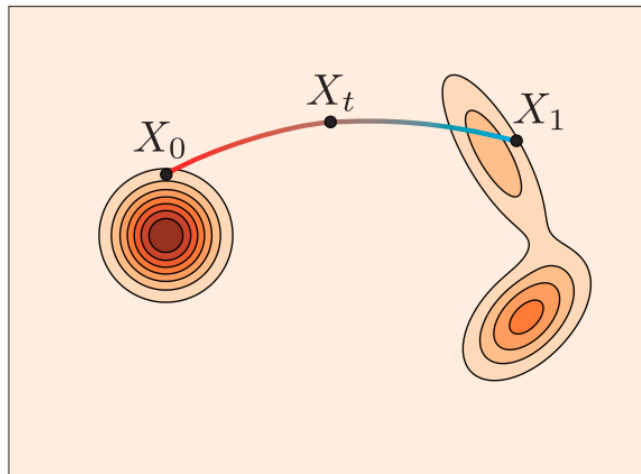
OpenVLA



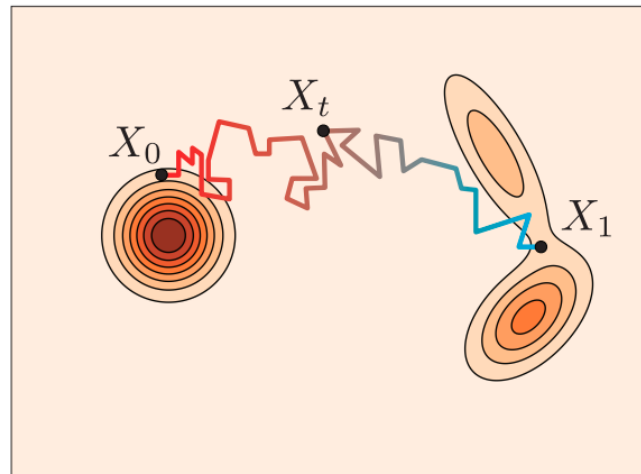
Preliminary) Diffusion



Preliminary) Flow Matching



(a) Flow



(b) Diffusion

$\pi 0.5$

$\pi 0.5$: a Vision-Language-Action Model with Open-World Generalization

Physical Intelligence

Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, Ury Zhilinsky

<https://pi.website/blog/pi05>

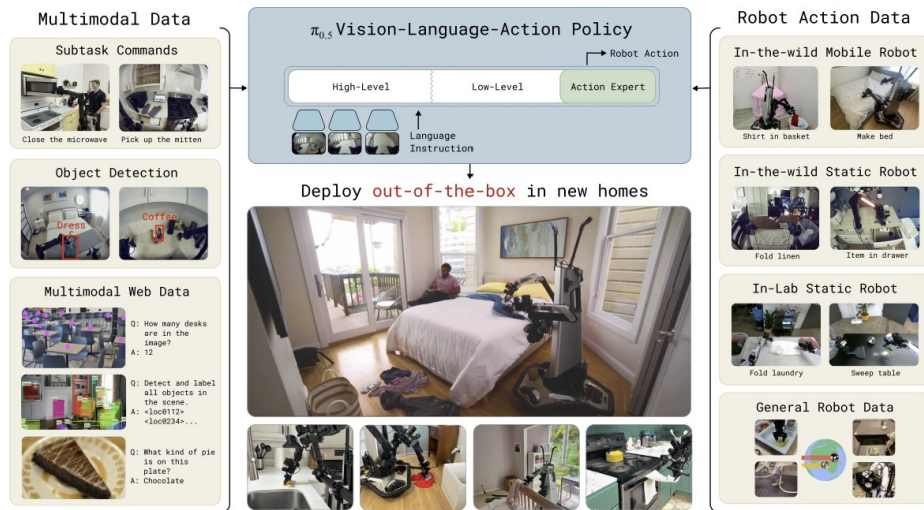


Fig. 1: The $\pi 0.5$ model transfers knowledge from a heterogeneous range of data sources, including other robots, high-level subtask prediction, verbal instructions, and data from the web, in order to enable broad generalization across environments and objects. $\pi 0.5$ can control a mobile manipulator to clean kitchens and bedrooms in new homes that were not present in the training data, performing complex multi-stage behaviors with durations of 10 to 15 minutes.

$\pi 0.5$

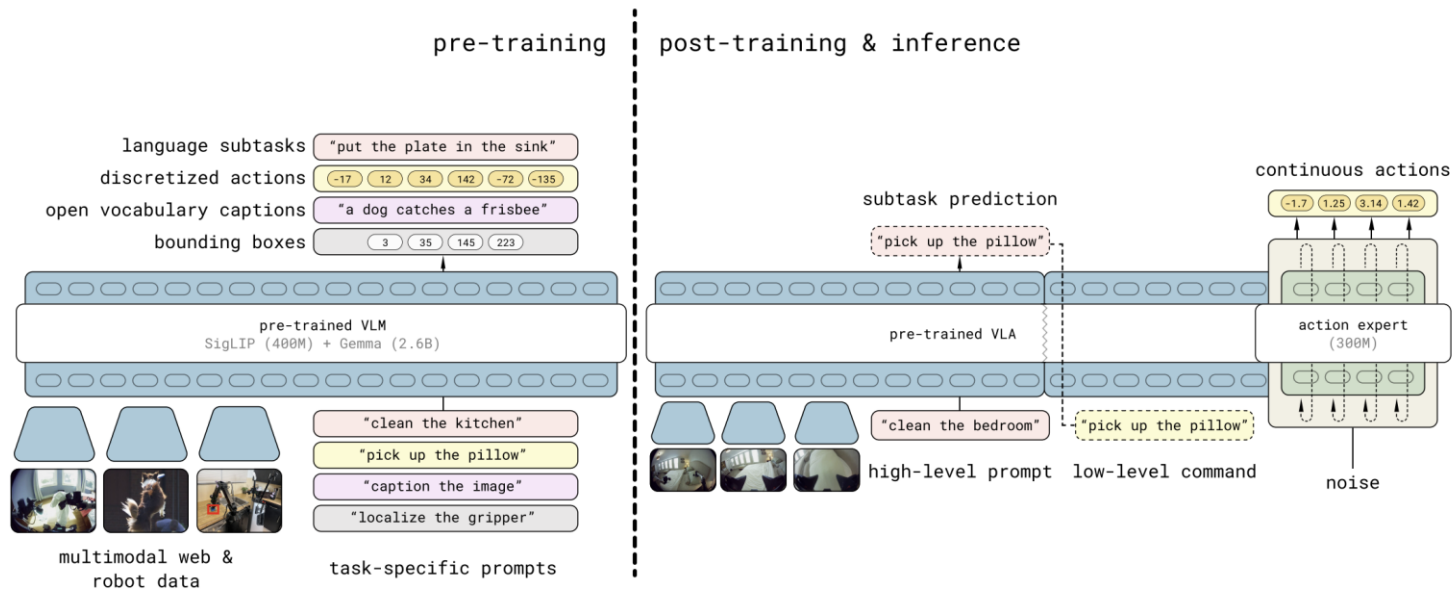
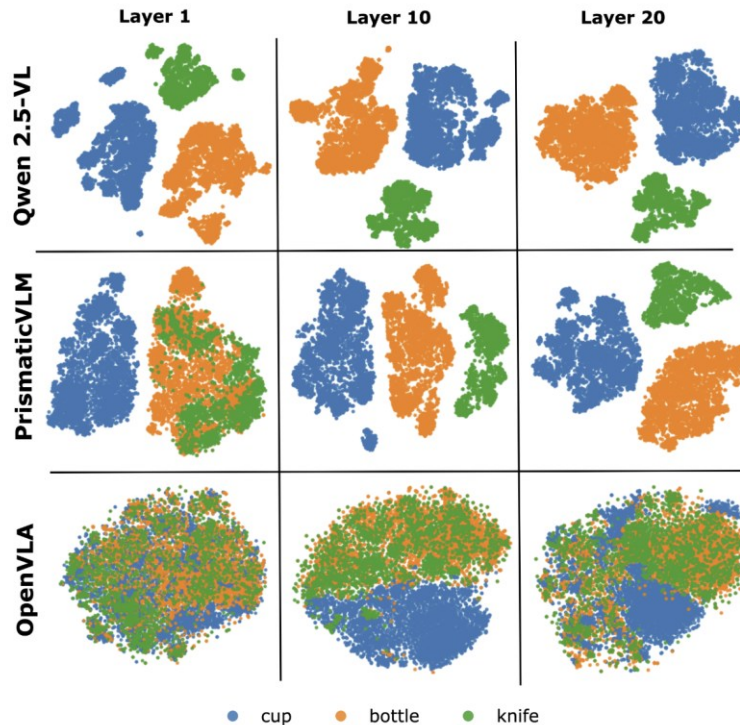


Fig. 3: **Model overview.** $\pi 0.5$ is trained in two stages. First, a pre-training stage combines all of the different data sources to produce an initial VLA with discrete tokens. This stage uses data from diverse robotic platforms, high-level semantic action prediction, and data from the web. Robotic data uses the FAST action tokenizer to represent actions as discrete tokens [64]. Second, a post-training stage specializes the model for low-level and high-level inferences for mobile manipulation, leveraging the most task-relevant data, including verbal instructions from human supervisors. This stage uses flow matching to represent the action distribution, enabling efficient real-time inference and the ability to represent fine-grained continuous action sequences. At inference time, the model first infers a high-level subtask, and then predicts the actions based on this subtask.

Action Expert를 따로 설계하는 이유 - Representation Collapse

Representation Collapse



t-SNE visualization of token embeddings. PrismaticVLM and Qwen2.5-VL maintain well-separated clusters for target objects, while OpenVLA shows strong overlap across classes, indicating that action fine-tuning leads to representation collapse.

LIBERO - Trajectory Memorization

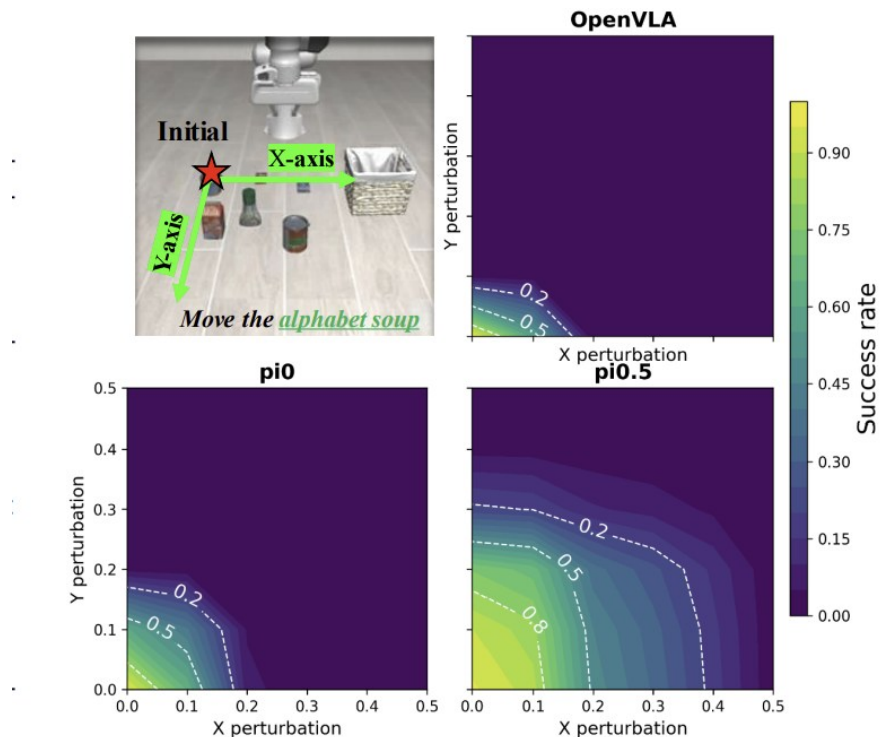
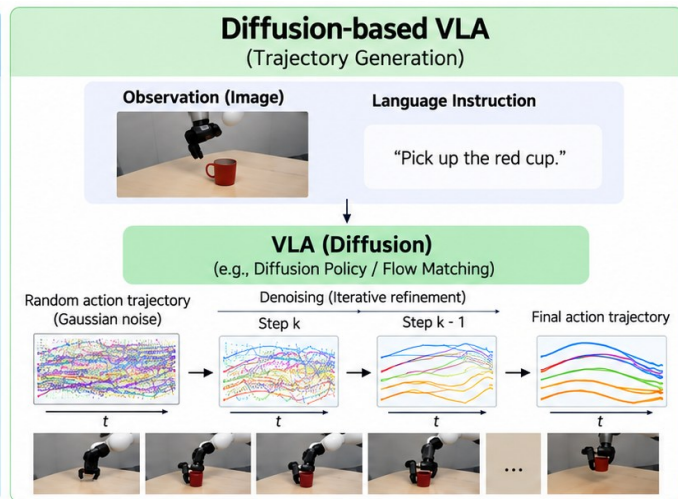
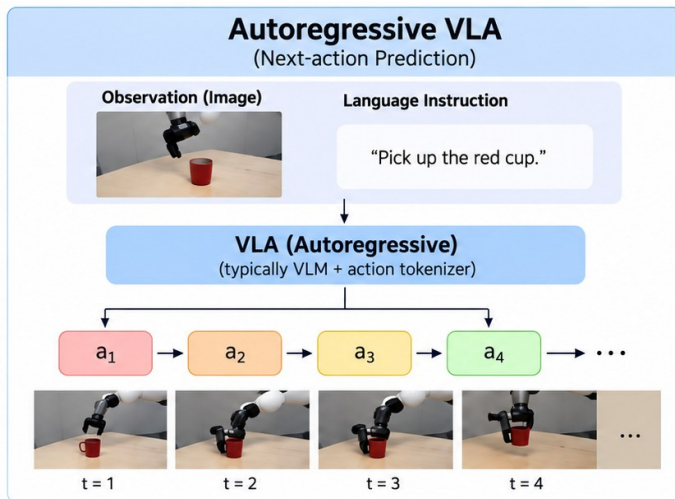


Figure 4: Success rates of OpenVLA, pi0, and pi0.5 under object position perturbations.

Training Recipe on Pi0.5

- **Diverse environments** — 100+ homes
- **Cross-embodiment data** — multiple robot platforms
- **High-level subtask prediction** — semantic task decomposition
- **Web & verbal instruction data** — broader visual and semantic knowledge

Autoregressive vs Diffusion



항목	Autoregressive VLA	Diffusion-based VLA
핵심 직관	LLM처럼 매 step에서 다음 action을 예측	전체 trajectory를 한 번에 두고 점진적으로 정제
장점	- 구조가 직관적이고 구현이 비교적 단순 - VLM/LLM backbone과 자연스럽게 결합 - action token prediction, instruction following에 강점	- multimodal action distribution 표현에 유리 - trajectory의 smoothness와 전체 일관성 확보에 강함 - continuous control에 잘 맞음
단점	- 순차 생성이라 horizon이 길수록 decoding 부담 증가 - 앞선 예측 오류가 뒤 step에 누적될 수 있음 - 연속 제어에서는 trajectory 전체 최적화가 어려울 수 있음	- iterative denoising으로 inference cost가 클 수 있음 - 학습/샘플링 구조가 더 복잡함 - step 수, noise schedule 등 설계 요소 영향이 큼
잘 맞는 경우	discrete action / action token 기반 정책	continuous robot control / manipulation

Action Manifold Learning VLA

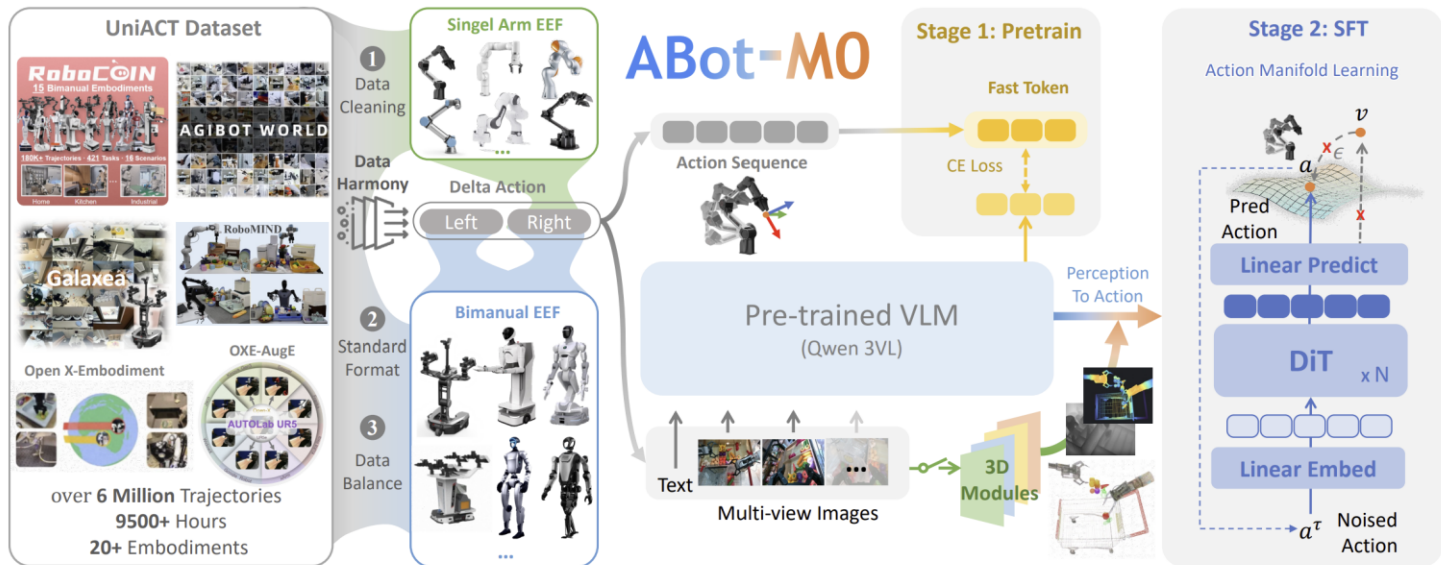


Figure 3 Model architecture of ABot-MO. We employ a two-component architecture consisting of a VLM and an action expert. In addition, we utilize action manifold learning to predict actions with two-stage training paradigm. We then carefully select VLM features and further introduce an optional 3D module and to enhance spatial reasoning.

World Action Models

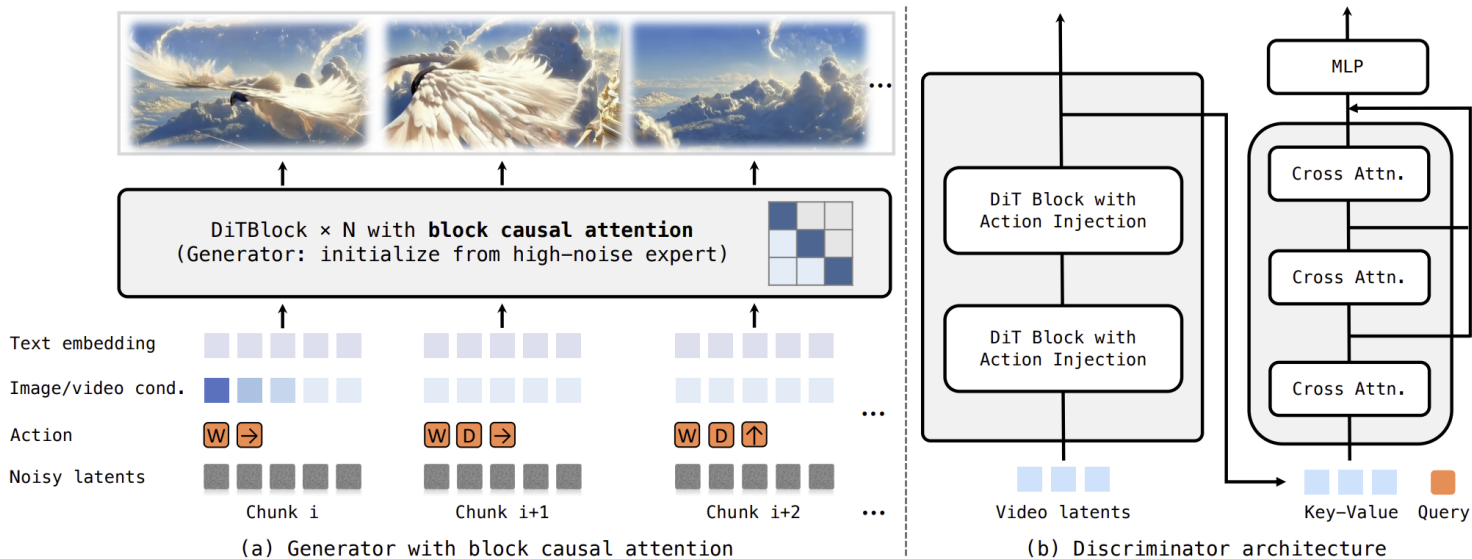


Figure 6. (a) Causal generator adaptation. To enable autoregressive streaming generation, we adapt the high-noise expert using **block causal attention**. This mechanism enforces global causality across chunks while maintaining local bidirectional consistency for efficient action-conditioned rollouts. **(b) Discriminator architecture.** For long-horizon training, we attach a GAN classification head $D(\cdot)$ to the fake score network features. This discriminator uses cross-attention to distinguish real from synthesized sequences to mitigate accumulative drift during distribution matching distillation.



ΠΟΗΝ